a Study design
Open data (no IRB)
Tahoe-100M single-cell drug atlas
379 drugs · 50 cancer cell lines (15 lung)
3 doses · 24 h · pseudobulk differential expression
Cardiac safety data
DICTrank · FAERS (openFDA) · ChEMBL hERG
iPSC-cardiomyocyte transcriptomes (Hansen 2024)
Cell-line drug sensitivity and baseline state
PRISM · GDSC · CTRP v2 · CCLE RNA-seq,
MS proteomics (CCLE, ProCan), RPPA
Analyses (D = discovery, C = confirmatory, T = v2 extension)
D2 Reproducibility and mechanism of action (positive controls)
D1 · C1 · T1-A · T2-H2 Cardiotoxicity vs chemistry; best-effort models; hERG
T2-H1 Drug identity: tumour lines vs iPSC-cardiomyocytes
D3 · D4 · C2 Pooled cell counts and response magnitude vs sensitivity
C3 · T2-E · T2-F · T1-D Tahoe response vs baseline RNA and protein, 3 screens
T1-A · T1-B · T1-C Best-effort models, LOO-bias simulation, RPPA targets
Pre-registered plan · 3 dated amendments · permutation nulls · robustness
b
Drugs per analysis
D1 DICTrank most vs no
D1 DICTrank most+less vs no
C1 FAERS signal
T2-H2 hERG pIC50
T2-H1 cardiomyocytes
C3 PRISM
C3 GDSC
T2-E CTRP
T2-F ProCan, PRISM
T2-F ProCan, GDSC
T2-F ProCan, CTRP
D2 mechanism classes
100
167
128
43
25
43
37
46
44
37
47
62
aqua = v2 extension
0
50
100
150
200
Number of drugs
c
Signature reproducibility
Different drugs (median 0.22)
Same drug (median 0.68)
Density
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
0.0
0.5
1.0
Correlation of mid- vs high-dose signatures
d
Mechanism recognition
Permutation null
Observed 0.32 (p = 0.0005)
Chance 0.09
Permutations
0
50
100
150
200
250
0.0
0.1
0.2
0.3
Balanced accuracy
(1-nearest neighbour, 11 classes)
e
Correct by mechanism class
DNA synth./repair
EGFR/ERBB
Multi-TK
Other TK
Microtubule
CDK
HDAC
JAK/STAT
MEK
RAF
Hedgehog
6/17
1/8
1/7
0/7
1/5
0/3
2/3
2/3
3/3
1/3
0/3
Drugs in class
Correctly recognised
0
10
20
30
Drugs
f
Mean signature similarity between mechanism classes
DNA synth./repair 0.19 0.19 0.16 0.12 0.24 0.11 0.17 0.22 0.18 0.21 0.13
EGFR/ERBB 0.19 0.21 0.19 0.17 0.22 0.15 0.10 0.21 0.29 0.29 0.15
Multi-TK 0.16 0.19 0.15 0.12 0.16 0.11 0.10 0.22 0.28 0.20 0.19
Other TK 0.12 0.17 0.12 0.10 0.16 0.10 0.13 0.25 0.06 0.05 0.10
Microtubule 0.24 0.22 0.16 0.16 0.33 0.14 0.18 0.38 0.15 0.27 0.22
CDK 0.11 0.15 0.11 0.10 0.14 -0.14 0.07 0.11 0.16 0.14 0.09
HDAC 0.17 0.10 0.10 0.13 0.18 0.07 0.39 0.36 0.03 -0.12 0.05
JAK/STAT 0.22 0.21 0.22 0.25 0.38 0.11 0.36 0.52 0.11 0.08 0.22
MEK 0.18 0.29 0.28 0.06 0.15 0.16 0.03 0.11 0.66 0.47 0.18
RAF 0.21 0.29 0.20 0.05 0.27 0.14 -0.12 0.08 0.47 0.74 0.21
Hedgehog 0.13 0.15 0.19 0.10 0.22 0.09 0.05 0.22 0.18 0.21 -0.05
DNA synth./repair
EGFR/ERBB
Multi-TK
Other TK
Microtubule
CDK
HDAC
JAK/STAT
MEK
RAF
Hedgehog
Mean Pearson r
0.4
0.2
0.0
−0.2
−0.4
Within-class (diagonal) mean r = 0.28 · between-class mean r = 0.17